

Scenario:

Option B: Mobile Payment Fraud Detection A mobile money service (like Payme or Click) processes millions of transactions monthly. They want an ML system that flags suspicious transactions in real time. They have historical transaction logs, user profiles, and a small set of confirmed fraud cases labeled by their security team.

Task 1: Problem Type

This is a binary classification problem, because the model must decide whether a transaction is fraudulent or not.

It is challenging due to class imbalance, since fraudulent transactions are rare compared to normal ones.

Task 2: Data Requirements

1. transaction_id — unique transaction identifier
2. amount — transaction amount
3. timestamp — time of transaction
4. sender_id — sender
5. receiver_id — receiver
6. transaction_type — type of transaction
7. device_info — device used
8. location — geographic location
9. account_age — age of account
10. transaction_frequency — number of transactions over time

Task 3: Evaluation Metric

I choose F1-score because it balances precision and recall.

- High precision reduces false alarms
- High recall catches more fraud

F1-score provides a balanced evaluation.

Task 4: Bias and Fairness Risks

Risks include:

- Rural users being flagged more often
- Low-income users being misclassified
- New users being considered suspicious
- Language-based bias

Task 5: Privacy Concerns

Collected data:

- financial transactions
- location
- device information
- behavior

Risks:

- data breaches
- identity theft
- misuse of personal data

Task 6: Recommendation

I recommend deploying the system with guardrails:

- human review
- regular evaluation of the model to detect and reduce bias
- a process that allows users to challenge or dispute the system's decision

Task 7: Reflection Questions

1. Yes, stricter laws like GDPR require stronger privacy protections and transparency.
2. Yes, but accuracy would decrease due to limited data.
3. Responsibility should lie with the company deploying the system.